Research Methods in Computer Science - COMP516-202223
Project Specification - Group 34
Multilingual Chatbot System as a Service

Teams	Scope	Name, Student ID	Name, Student ID	Name, Student ID
Team A	System Design and Implementation	Kshitij, 201668660	Nishanth, 201629072	
Team B	Translation Systems and UI/UX	Robin, 201685997	Kwonmo Kang, 201679715	CHI YEH, LI, 201623581
Team C	Chatbot Systems	Vijeth, 201664207	Kailash, 201682735	

Supervisor

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Project Description
A.Technical Summary

1. The platform will enable several deployment methods and will be hosted on a cloud platform. The software will be trained by data scientists utilising a bilingual lookup cache. As a result, we will start by discussing a few frequently spoken languages before moving on.

2. Untranslatable words will have their image-mapping URLs kept in the bilingual lookup cache [1]. With our caching, we would desire eventual consistency. If the cache fails, we would also need the identical data kept on the database.

3. Our system supports global lookups for individual organisations. For similar business sectors as seen in [2], we can have certain mutuals (HealthCare A and B can have some mutual mappings and some organisation-specific mappings).

4. Organisations will have access to translation software and a chatbot[3] with a built-in training interface.

5. We will make a B2C translation API available. To utilise our APIs, the users are required to create a user account and client credentials. A B2C customer API and a feedback API will also be available on the platform.

B.Non-Technical Summary

1. Translation systems will be applied to chat assistance for various business sectors, which has been demonstrated in [4, Ch. 1, pp. 1-21].

2. We can program our chatbots or even use pre-built ones that can be offered as a service.

3. A business-to-consumer element will be a part of our system. For those who want to undertake translations for their projects or small-scale usage, a requirement-specific based service will be made available.

Aim

To develop B2B and B2C business models for translation systems. AI chatbots have become more prevalent in our daily lives. We will use the knowledge gained from AI-chatbot integrations to build a stronger, more robust AI chatbot that can be sold separately.

Objectives

1. 90% test coverage for our code-base.
2. Sub 1s translations of each input.
3. Throughput of 10K requests per minute for our B2C exposed API.
4. More than 95% confidence score for translations on at least 97% of inputs.
5. 4 9s availability under Service Level Agreement [5].

Key Literature and Background Reading

Our plan from the start was to do an asynchronous attempt at the project specification by taking on all three important aspects of our work. These are the breakdowns of the readings.
1. Translation System: Kshitij, Kang, Chi-yeh, and Robin
2. Chatbots: Nishanth, Kailash, and Vijeth
3. System Design: Kshitij and Nishanth.

For the AI chatbot, we read about chatbots in various business contexts to help us to train a generic chatbot for our translation system. According to the study [6][7], the fundamental technology behind chatbots nowadays is connected to our line of work. Additionally, according to "Multimedia Chat for Help Desks"[8],the conversation is between the expert/agent and non-expert user.

As demonstrated in [9]-[11], the use of intermediate languages and bilingual dictionaries to improve translation quality. Used this paper to develop our translation service.

As seen in [9] helped us focus on an important point that people forget about, the conversation is between an expert or agent, and the user who might not be an expert.

As previously indicated [10], [12], and [13], it enabled us to realise the value of global lookups for different types of organisations and local lookups for bilingual dictionaries, for non-translatable words, inspiration was also taken from the machine translator.

For AI Chatbot, it was important for us to read about chatbots in different business verticals. For instance, [14]-[16] have a crucial component of trying to extract information about emotion from subtext.

Our aim is to collate information that will be applicable to different types of applications, that will help us to train a generic chatbot and put it up as an add-on for our translation system. The paper that is mostly related to our work is Khanna et al. research[7], [17] and [18].

Development and Implementation Summary

We are going to use Golang as our programming language for the API service. Python is going to be used for translation services and AI-chatbots.

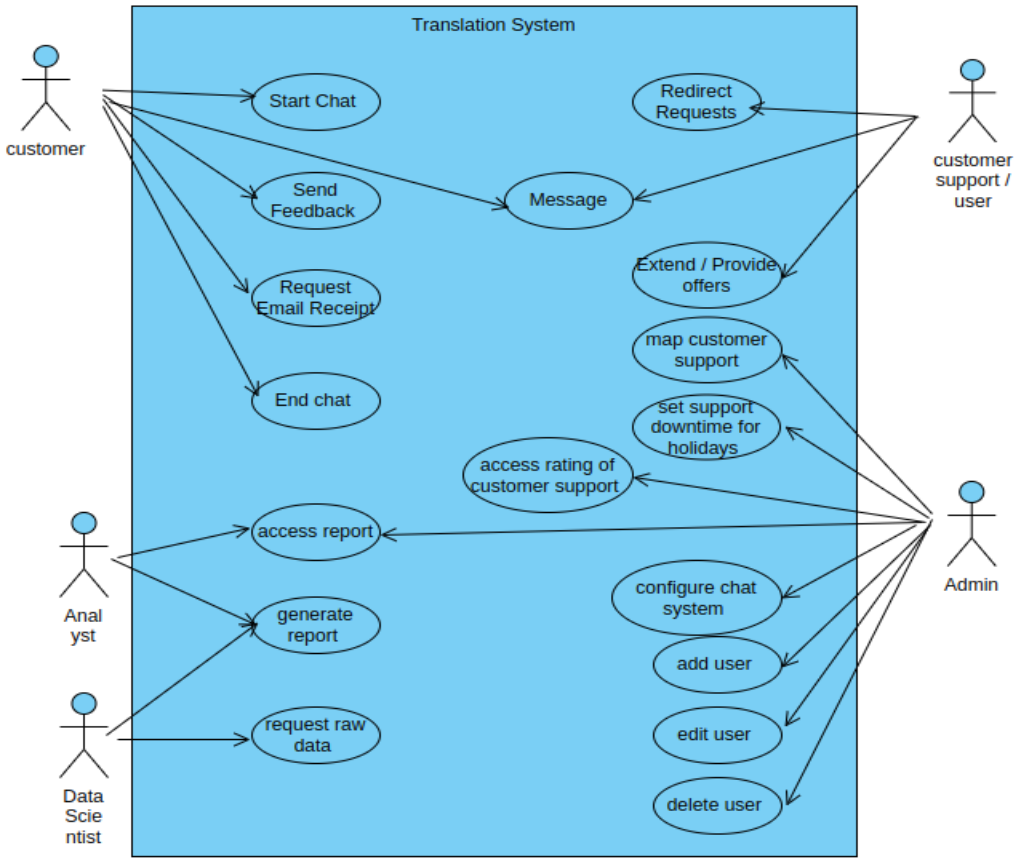
We are using **Golang** for our API service because:

1. It's strictly typed.
2. Has a robust internal package for testing [19] and benchmarking
3. Easy to implement a domain-driven design that we are expected to use.

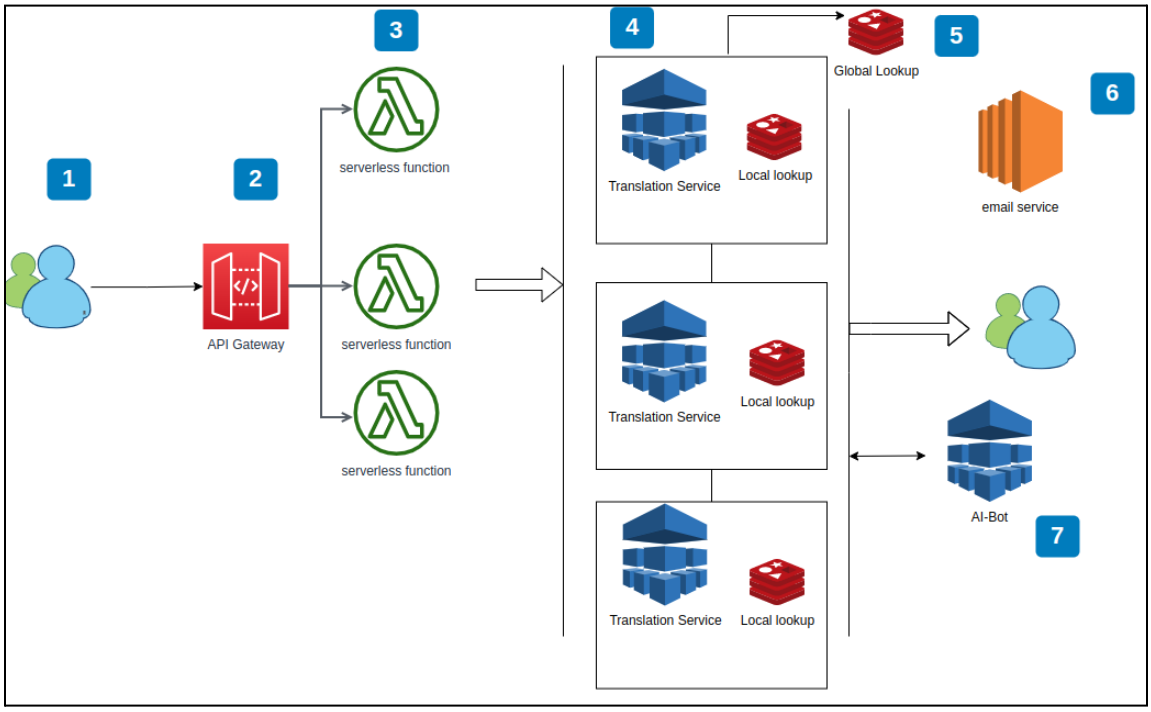
We are using **Python** for our translation and ai-chatbot services because:

1. Python is already well-established in the ML community
2. Being simplistic in nature and sequentially fast, the learning curve is not steep and hence the main focus can be put on implementation.
3. The Machine Learning Libraries in python[20] are well maintained and up to date by the python open-source community.

Below is the **use-case** diagram which shows who are the users of our translation system and what actions they are going to perform.



High-Level Design



Components for Software-as-a-Service:

API Request
The endpoint that the customer will be using to communicate with our system.

API Gateway
API Gateway will act as a load balance. It will have basic security features so that the headers are not lost or misused in transition.

Serverless Function - *AWS LAMBDA [21, Ch.6]*
It will validate the request, refine the request, and hit the translation service for the translations.

Translation Service with local lookups
The local lookups will have bi-lingual mappings between English and another language. Each instance may have one or more than one bi-lingual mappings. These bilingual mappings will be accompanied by image references. These are the words that don't have direct translations.

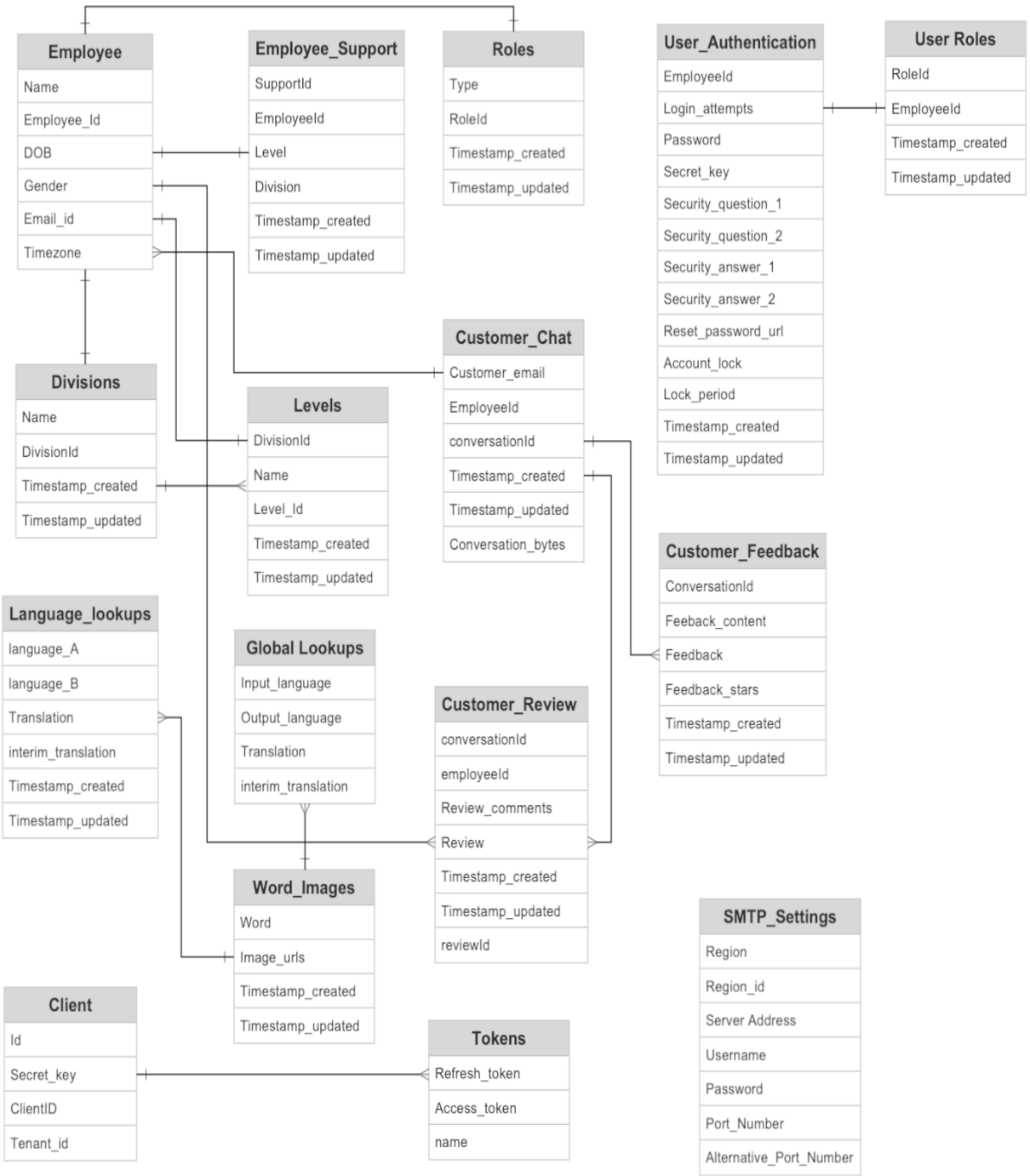
Global lookups are there for industry-specific lookups with image references.

Email service for sending the transcripts to the end-user.

Integratable AI chatbots (Native or Client's add-on).

Entity-Relationship Diagram

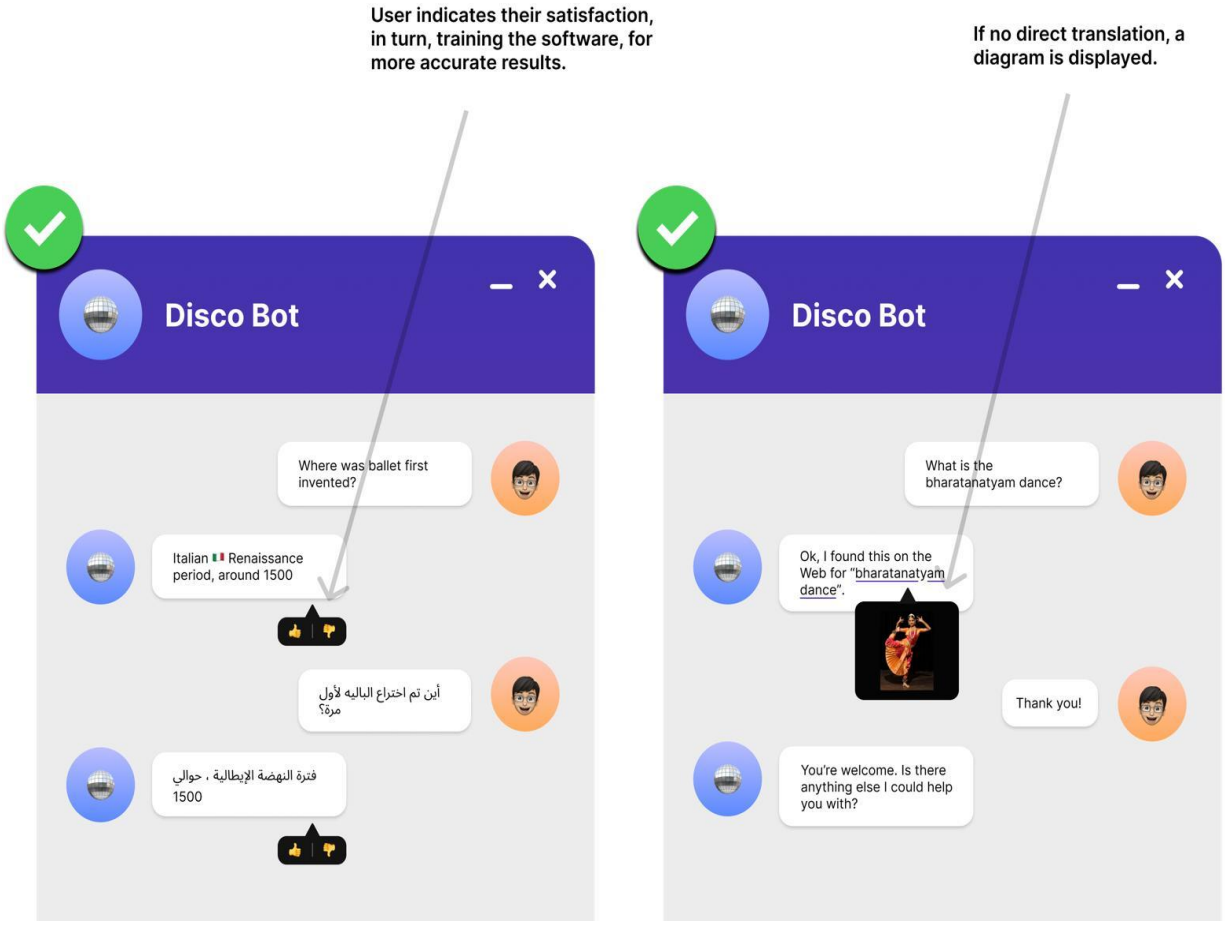
A low-level description of our database for the administration, translation and support-mapping is illustrated below:



User Interface-Mockup

For a successful chatbot interface, these are the things we prioritised:

1. Abstraction: Maximum outputs with minimum user inputs.
2. Clean UI.
3. Colours and fonts accessible to old and colour-blind people.
4. Later iterations, audio translations, and speech-to-text handling, for the visually impaired.



Key takeaways from the mock UI above:

1. 👍 (“Thumbs up”): Positive feedback for the translation.
2. 👎 (“Thumbs down”): Negative feedback on the translation.
3. Chatbot auto-recognises the language and changes its response language accordingly.
4. Image reference for non-translatable words.

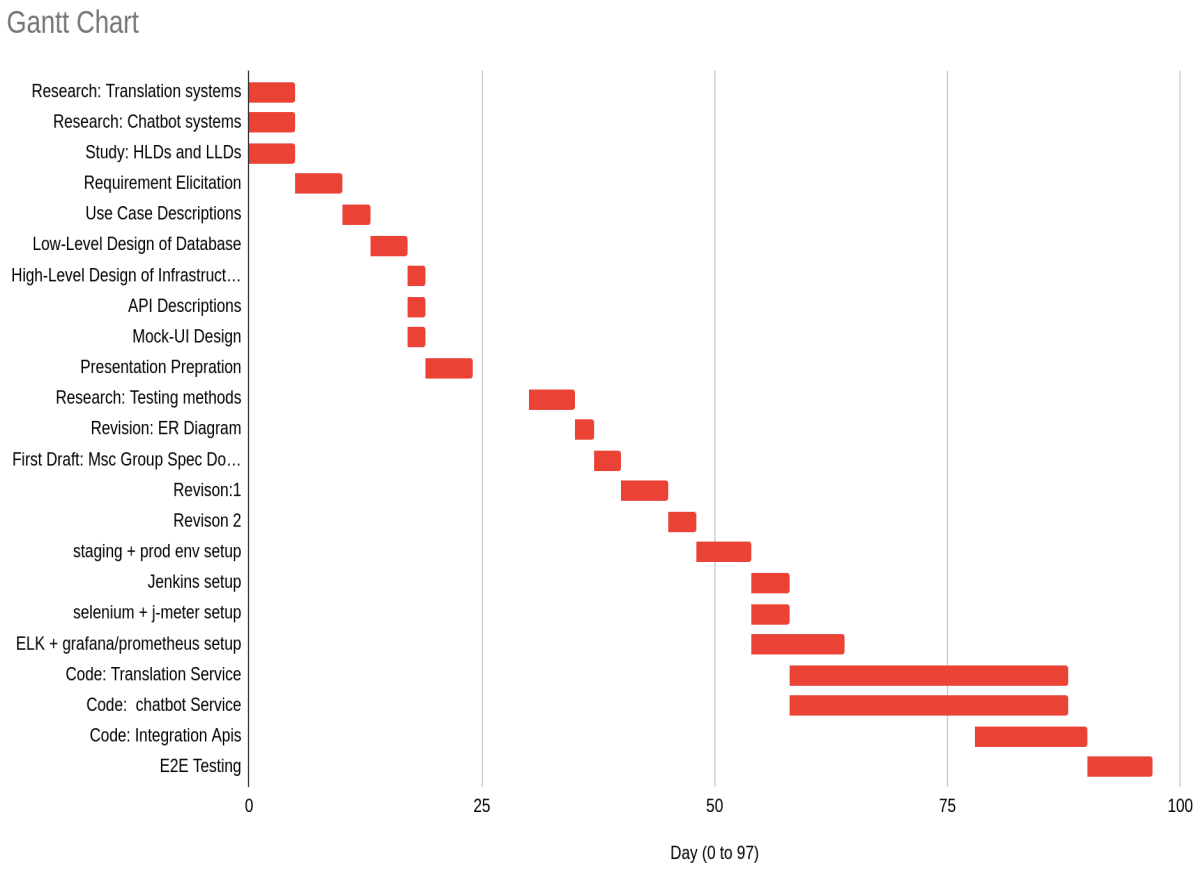
Project Plan

The group was divided into three teams. These are as follows:

1. Team A (Kshitij and Nishanth)
   Responsibilities include:
   a. Requirement elicitation and converting them into logical variables.
   b. High-Level Design, Low-Level Design, ER Diagrams and Test Planning.

2. Team B (Vijeth And Kailash)
   Responsibilities include:
   a. Design and implementation of AI-Chatbot systems.

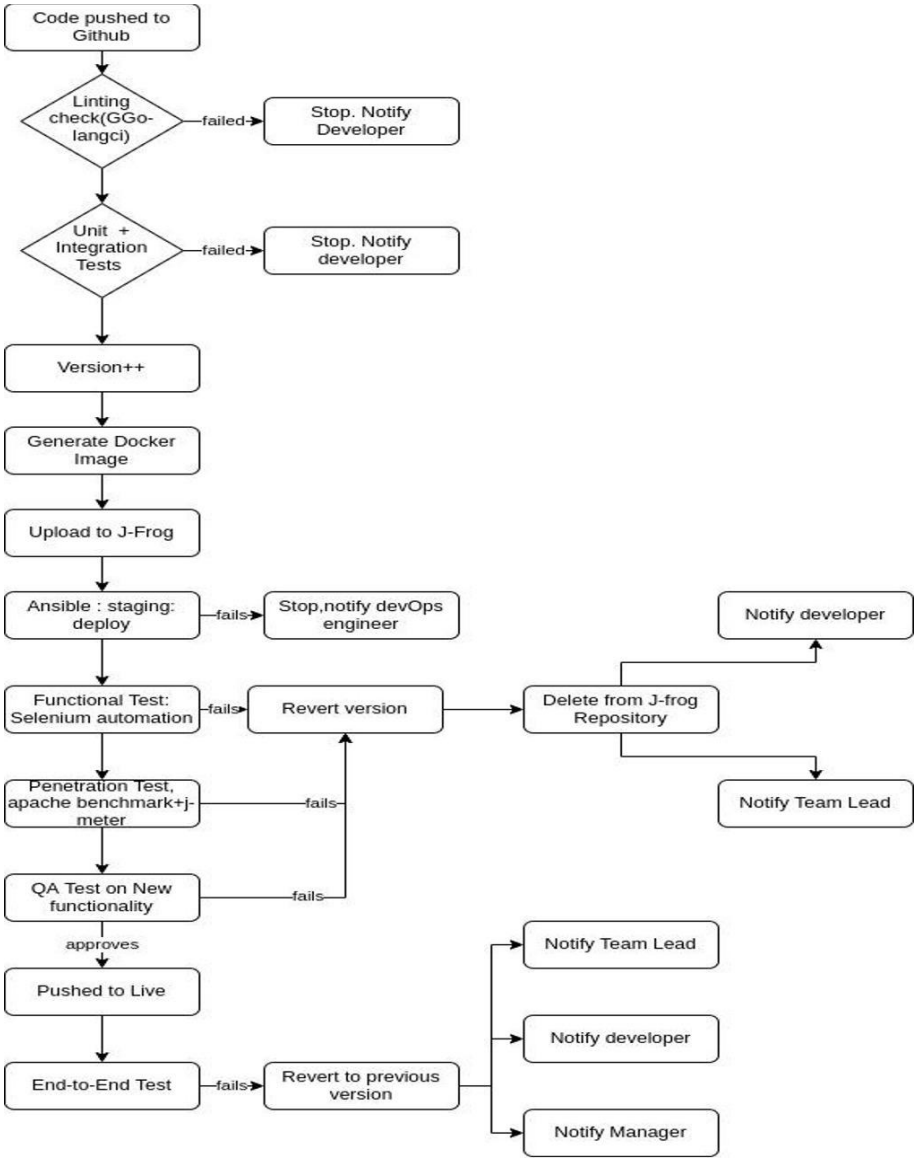
3. Team C (Chi-yeh, Kang, Robin)
   Responsibilities include:
   a. Design and Implementation of Translation Service.
   b. UI/UX Design.



Testing

We will have to build a Continuous Integration and Continuous Deployment pipeline for our application. We are doing agile development, so the code will be pushed to live frequently.
We will be using Jenkins as our CI-CD pipeline because expertise in the group is already there. Another option could have been GitHub actions as we are using GitHub as our code repository, but deployment and functional testing on serverless would be slightly different.
Developers will work in the development branch, and CI-CD will work on staging and live branches. Developers can only push till the staging branch, the manager has to manually accept pull requests for the code to be pushed to the live branch.

These are the steps below that are involved in the CI-CD process.



Evaluation

To fully test our entire multilingual chatbot system and uncover any bugs in the created product, we will employ a team of testers. It helps in the detection of errors, missing requirements, and problems by determining whether the predicted outcomes can match the anticipated results. If our system passes the prerequisites checklist, it will be able to accomplish the chatbot's goals. The software testing team will issue the certification attesting to the reliability of our product before it is supplied to the person (B2C) or organisation (B2B). We will review the data from the recently delivered chatbot system after 30 days have passed since our product was delivered in order to analyse and evaluate the success rate of our translations of our delivered chatbot system. We will check for a minimum confidence score of over 95% confidence score on our machine-learning translation algorithm for at least 97% of inputs. Throughput of 10000 requests per minute for the exposed translation API for the B2C model and evaluate the duration of our multilingual chatbot’s uptime (availability) which should be “four nines” (4.38 minutes downtime per month) under the Service-Level Agreement (SLA).

Ethical Considerations

Our chatbot was developed by a diverse team, improving performance through each iteration. Anti-bias training and AI/ML training can identify and eliminate biases. The chatbot's training data is labelled accordingly and annotated to ensure fairness. Transparency in decision-making [6] enhances trust and legitimacy in the chatbot system. Allowing developers the opportunity to gain customer insights safely and prevent ethical issues.

Risks and Contingency Plans

1. Technical contingency plans for our multilingual chatbot will include:
   a. Setting up routine bi-weekly check-ins for evaluating the performance.
   b. If the success rate doesn't go up within the first few weeks, we will train our models with open-source datasets for translation as well as chatbot service.
2. We will use the STRIDE [22] (Spoofing identity, Tampering the data, Repudiation threats, Information disclosure, Denial of service and Elevation of privileges) model and SDL (Security Development Lifecycle) . We will outsource LMS to help staff get training in email-securities. As phishing is the leading cause of data breaches, we will equip our mailbox with post-delivery email security systems.
3. In case a computer rack for our on-premise services or cloud services goes down, we will ensure that the same service is hosted on several racks/ regions to have high availability.
4. We will be using Cloudflare DDoS protection to protect our exposed endpoints or interfaces from DDoS attacks.
5. We will adhere to the rules defined under GDPR and The Data Protection Act 2018 (UK version of GDPR)[23] to prevent Legal or Reputational Risks.

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